Introduction to the ab initio nuclear many-body problem

Lecture 4: *Quantum Monte Carlo as an ab initio method*

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Contents

1 Quantum Monte Carlo Methods for many-body quantum mechanics

2 The Quantum Monte Carlo Family of approaches

3 Variational Monte Carlo (VMC)

3.1 Introduction

Variational Monte Carlo (VMC) is typically formulated in coordinate space. We will first set up the appropriate language to describe systems in coordinate space, and use it to express the Hamiltonian. Then we will discuss the concept of a trial wavefunction, that is, our best guess for the Hamiltonian's ground state. Finally we will reformulate Schröndiger's equation in a variational form.

3.2 The QMC-*esque* coordinate space language

A system of N particles in d dimensions is described using Nd-dimensional vector of all positions of all particles

$$\mathbf{X} = \left(r_{11}, r_{12}, \ldots, r_{1d}, \ldots r_{N1}, r_{N2}, \ldots, r_{Nd}\right)^T \quad , \tag{1}$$

where r_{ij} is the j-th coordinate of the i-th particle, e.g., the second (i.e., y coordinate) of the 6th particle is r_{62}. For example, for 2 particles in 3 dimensions ($N = 2,\ d = 3$),

$$\mathbf{X} = \left(r_{11}, r_{12}, r_{13}, r_{21}, r_{22}, r_{23}, \right)^T \quad . \tag{2}$$

We are after the *ground state energy*of the system, E_0. This is usually the point of departure for any other observable. Variational Monte Carlo provides an upper limit to E_0.

3.3 The Hamiltonian

The many-body Hamiltonian can be written in coordinate space

$$\hat{H} = \sum_{i=1}^{N} \hat{K}_i + \sum_{i<j} \hat{V}_{ij}^{(2)} + \sum_{i<j<k=1}^{N} \hat{V}_{ijk}^{(3)} + \cdots \tag{3}$$

$$\hat{H} = \sum_{i=1}^{N} \left(-\frac{\hbar^2}{2m}\boldsymbol{\nabla}_i^2\right) + \sum_{i<j=1}^{N} V^{(2)}\left(\mathbf{r}_i, \mathbf{r}_j\right) + \sum_{i<j<k=1}^{N} V^{(3)}\left(\mathbf{r}_i, \mathbf{r}_j\, \mathbf{r}_k\right) + \cdots \tag{4}$$

$$\hat{H} = \sum_{i=1}^{N} \left(-\frac{\hbar^2}{2m}\boldsymbol{\nabla}_i^2\right) + \sum_{i<j=1}^{N} V^{(2)}\left(\mathbf{r}_i, \mathbf{r}_j\right) + \sum_{i<j<k=1}^{N} V^{(3)}\left(\mathbf{r}_i, \mathbf{r}_j\, \mathbf{r}_k\right) + \cdots \tag{5}$$

$$\tag{6}$$

where $\boldsymbol{\nabla}_i$ is the vector $(\partial_{r_{i1}}, \partial_{r_{i2}}, \ldots, \partial_{r_{id}})$ acting on the coordinates of particle i. The superscript of the operators $V^{(n)}$ signifies an n-body interaction: in QMC methods dealing with $N > 2$-body interactions is relatively straightforward, which is a relief for the study of nuclear systems (see Lecture 3).

Why is there $i < j$ in the sum of two-body interactions?

The form of the Hamiltonian, e.g., in Eq. (5), guides the choice for a trial wavefunction. Inspired by ideas from perturbation theory, we could write the Hamiltonian as $\hat{H} = \hat{H}_0 + \hat{H}_1$, where $\hat{H}_0$'s ground state can be easily determined, and use that as a trial wave-function. Alternatively, we could use a phenomenological approximation to the ground state of $\hat{H}$ for a trial wavefuntion. More details, below.

Finally, to pose the problem well, we need to specify the boundary conditions that are applied to the particles. Note that even though we will not be solving and differential equations directly, the boundary conditions will enter our calculations.

Identify where the boundary conditions enter in the VMC formulation

3.4 Trial wavefunction

The trial wavefunction is our best guess for how the ground-state of Eq. (5) looks like. *One cannot prove within VMC that the guess is exactly correct.* However, due to the variational principle, we can guarantee that the energy calculated with it is an upper bound of the ground-state energy. This creates a practical strategy: one can keep updating the guesses until the calculated energy doesn't change. As we will see later, this can be automatized to an extent. In what follows we'll go over some standard choices for trial-wavefunctions.

A slater determinant As already mentioned above a standard choice for the ground-state's form is the Slater Determinant (SD); see Lecture 1. This corresponds to perturbation-theory inspired strategy mentioned above, i.e., splitting into two parts $\hat{H} = \hat{H}_0 + \hat{H}_1$, with $\hat{H}_0 = \hat{K}$ and using eigenstates of $\hat{H}_0$ as trial states:

$$\Phi_0^{(\mathrm{SD})}\left(\mathbf{X};\mathbf{k}_1,\mathbf{k}_2,\ldots,\mathbf{k}_N\right) = \mathcal{A}[\phi_{\mathbf{k}_1}(\mathbf{r}_1),\phi_{\mathbf{k}_2}(\mathbf{r}_2),\ldots,\phi_{\mathbf{k}_N}(\mathbf{r}_N)] = \det\begin{pmatrix} \phi_{\mathbf{k}_1}(\mathbf{r}_1) & \phi_{\mathbf{k}_1}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_1}(\mathbf{r}_N) \\ \phi_{\mathbf{k}_2}(\mathbf{r}_1) & \phi_{\mathbf{k}_2}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_2}(\mathbf{r}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{\mathbf{k}_N}(\mathbf{r}_1) & \phi_{\mathbf{k}_N}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_N}(\mathbf{r}_N) \end{pmatrix} \tag{7}$$

where $\phi_{\mathbf{k}_i}(\mathbf{r}_j)$ is an eigenstate of K with with energy $E_i = \hbar^2 k_i^2/2m$ evaluated at the coordinate of particle j:

$$K\phi_{\mathbf{k}_i}(\mathbf{r}) = \frac{\hbar^2}{2m}k_i^2\phi_{\mathbf{k}_i}(\mathbf{r}) \tag{8}$$

In the case of a spin-balanced system of N particles, that is, with $N/2$ particles per species(spin-up and spin-down)[1], we can generalize this trial wavefunction to the product of two SDs:

$$\Psi_0^{(\mathrm{SD})}\left(\mathbf{X};\mathbf{k}_1,\mathbf{k}_2,\ldots,\mathbf{k}_{N/2},\mathbf{k}_{1'},\mathbf{k}_{2'},\ldots,\mathbf{k}_{N/2'}\right) = \Phi_0^{(\mathrm{SD})}\left(\mathbf{k}_1,\mathbf{k}_2,\ldots,\mathbf{k}_{N/2}\right)\Phi_0^{(\mathrm{SD})}\left(\mathbf{k}_{1'},\mathbf{k}_{2'},\ldots,\mathbf{k}_{N/2'}\right) \tag{9}$$

$$= \mathcal{A}[\phi_{\mathbf{k}_{1'}}(\mathbf{r}_{1'}),\phi_{\mathbf{k}_{2'}}(\mathbf{r}_{2'}),\ldots,\phi_{\mathbf{k}_{N'/2}}(\mathbf{r}_{N'/2})]\mathcal{A}[\phi_{\mathbf{k}_1}(\mathbf{r}_1),\phi_{\mathbf{k}_2}(\mathbf{r}_2),\ldots,\phi_{\mathbf{k}_{N/2}}(\mathbf{r}_{N/2})] \tag{10}$$

$$= \det\begin{pmatrix} \phi_{\mathbf{k}_1}(\mathbf{r}_1) & \phi_{\mathbf{k}_1}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_1}(\mathbf{r}_N) \\ \phi_{\mathbf{k}_2}(\mathbf{r}_1) & \phi_{\mathbf{k}_2}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_2}(\mathbf{r}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{\mathbf{k}_N}(\mathbf{r}_1) & \phi_{\mathbf{k}_N}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_N}(\mathbf{r}_N) \end{pmatrix} \det\begin{pmatrix} \phi_{\mathbf{k}_{1'}}(\mathbf{r}_{1'}) & \phi_{\mathbf{k}_{1'}}(\mathbf{r}_{2'}) & \cdots & \phi_{\mathbf{k}_{1'}}(\mathbf{r}_{N'}) \\ \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{1'}) & \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{2'}) & \cdots & \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{N'}) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{\mathbf{k}_{N'}}(\mathbf{r}_{1'}) & \phi_{\mathbf{k}_{N'}}(\mathbf{r}_{2'}) & \cdots & \phi_{\mathbf{k}_{N'}}(\mathbf{r}_{N'}) \end{pmatrix} \tag{11}$$

A BCS determinant A state with the lowest rang of many-particle correlations is the BCS state which can be written in the determinant form (see Lecture 1&2)

$$\Psi_0^{(\mathrm{BCS})}\left(\mathbf{X};\varphi\right) = \mathcal{A}[\varphi(\mathbf{r}_1,\mathbf{r}_2),\varphi(\mathbf{r}_3,\mathbf{r}_4),\ldots,\varphi(\mathbf{r}_{N-1},\mathbf{r}_N))] = \det\begin{pmatrix} \varphi(\mathbf{r}_1,\mathbf{r}_{1'}) & \varphi(\mathbf{r}_1,\mathbf{r}_{2'}) & \cdots & \varphi(\mathbf{r}_1,\mathbf{r}_{N'/2}) \\ \varphi(\mathbf{r}_2,\mathbf{r}_{1'}) & \varphi(\mathbf{r}_2,\mathbf{r}_{2'}) & \cdots & \varphi(\mathbf{r}_2,\mathbf{r}_{N'/2}) \\ \vdots & \ddots & \vdots & \\ \varphi(\mathbf{r}_{N/2},\mathbf{r}_{1'}) & \varphi(\mathbf{r}_{N/2},\mathbf{r}_{2'}) & \cdots & \varphi(\mathbf{r}_{N/2},\mathbf{r}_{N'/2}) \end{pmatrix} \tag{12}$$

[1]In the absence of spin-spin interactions, no spin-flips can occur and it is common to refer to each spin population as "species".

which depends functionall on the *pair wavefunctions*,

$$\varphi(\mathbf{r}_i, \mathbf{r}_{j'}; \{c_n\}, t) = \sum_n c_n \phi_{\mathbf{k}_n}(\mathbf{r}_i)\phi_{t(\mathbf{k}_n)}(\mathbf{r}_{j'}) \tag{13}$$

which is in turned defined by the pairing amplitudes c_n and the *pairing scheme* t which pairs single-particle state mfk to state $t(\mathbf{k})$. Note that instead of pairing momentum states one could also pair linear combination of them in the generalized pair wavefunction

$$\tilde{\varphi}(\mathbf{r}_i, \mathbf{r}_{j'}; \{c_n\}, t) = \sum_n \tilde{c}_n g_n(\mathbf{r}_i) g_{\tilde{t}(n)}(\mathbf{r}_{j'}) \tag{14}$$

where

$$g_n(\mathbf{r}) = \sum_m a_{nm}\phi_{\mathbf{k}_m}(\mathbf{r}) \tag{15}$$

This is more similar to the HFB state discussed in Lecture 2.

For example, in the simplest case of the singlet-pairing encountered in terrestial superconductors, the BCS pair state in Eq. (13) is a good phenomenological model with the pairing scheme t corresponding to the time-reversal operator and the pairing amplitudes from BCS theory,

$$t(\mathbf{k}) = -\mathbf{k}\ , \quad c_n = \frac{v(\mathbf{k})}{u(\mathbf{k})} \tag{16}$$

3.5 Recasting Schrondiger's Equation: the Rayleigh-Ritz principle

3.5.1 The general formulation

At the heart of VMC lies the fact that the expectation value of the Hamiltonian at *any* trial state ψ_T is an upper bound to the ground state. This is a general result that holds for any ψ_T, but, of course, its practical value is seen when ψ_T is chosen "close" to the ground state and its energy becomes a tight bound on the ground state energy. What is meant by "close" here will be quantified in this section to mean "large overlap in Hilbert space". Furthermore, we will validate the assumption that a the "closer" a trial state is to the ground state, the tighter the energy bound.

First we define the basis spanned by eigenstates of the many-body Hamiltonian in Eq. (5), $|\Phi_n\rangle$, that is,

$$H|\Phi_n\rangle = E_n|\Phi_n\rangle\ .$$

Even though we don't know much about $|\Phi_n\rangle$ (after all our goal here is to acquire access to $|\Phi_0\rangle$, or at least E_0), we will use that their eigenstates of H, their orthonormality, $\langle\Phi_n|\Phi_m\rangle = delta_{nm}$ and their completeness, i.e., that they form a basis, for a proof. Our trial state can be expressed as

$$|\Psi_T\rangle = \sum_{n=0}^{\infty} a_n|\Phi_n\rangle\ , \quad \text{with } a_n = \langle\Phi_n|\Psi_T\rangle \tag{17}$$

Note that we don't assume that $|\psi_T\rangle$ is normalized. The so-called variational energy E_V is just the normalized expectation value of the Hamiltonian at $|\Psi_T\rangle$,

$$E_V = \frac{\langle\Psi_T|H|\Psi_T\rangle}{\langle\Psi_T|\Psi_T\rangle} = \frac{\sum_{n'n} a^\star_{n'}a_n\langle\Phi_{n'}|H|\Phi_n\rangle}{\sum_{n'n} a^\star_{n'}a_n\langle\Phi_{n'}|\Phi_n\rangle} = \frac{\sum_n |a_n|^2 E_n}{\sum_n |a_n|^2} = E_0 + \frac{\sum_n |a_n|^2(E_n - E_0)}{\sum_n |a_n|^2} \geq E_0 \tag{18}$$

where to prove the bound we have used that $E_0 < E_n$ for every n. This result also allows us to use the variational energy to quantify when a state is "close" to the ground state. To see this more concretely, let's consider our trial state to be normalized $\langle\Psi_T|\Psi_T\rangle = 1$, or $\sum_n |a_n|^2 = 1$ (equivalently we could define $\tilde{a}_n = a_n/\sqrt{\sum_n |a_n|^2}$), such that the bound we just proved becomes

$$E_V = E_0 + \sum_n |a_n|^2 (E_n - E_0) \tag{19}$$

We will show that a small variation away from the ground state translates to a small shift in energy. Indeed, if our trial state is almost equal to the ground state, with only a small admixture of a state $|\Phi_1\rangle$, then $a_1 = \eta$, $a_0 = \sqrt{1-\eta^2}$, and $a_n = 0$ otherwise, where we are implying that η is a small number. This means,

$$|\Psi_T\rangle = \sum_{n=0}^{\infty} a_n |\Phi_n\rangle = \sqrt{1-\eta^2}\,|\Phi_0\rangle + \eta\,|\Phi_1\rangle \tag{20}$$

and the bound becomes

$$E_V = E_0 + \sum_n |a_n|^2 (E_n - E_0) = E_0 + \eta^2 (E_1 - E_0) \tag{21}$$

In words, as the excited-state components of the trial state decrease *linearly* the variational energy approaches the ground state energy *quadratically*. So, the variational energy can quantify the distance of the trial state from the ground state. This provides the motivation behind VMC: stepping through a Hilbert space in a direction that decreases the variational energy is a path that leads to the ground state. If one parametrizes such a walk with a set of parameters $\boldsymbol{\alpha}_i$, such that $|\Psi_T(\boldsymbol{\alpha}_i\rangle$ is a state along this walk, finding the best approximation to the ground state is reduced to a standard *multi-dimensional* minimization problem that involves minimizing the function

$$E_V(\boldsymbol{\alpha}) = \frac{\langle\Psi_T(\boldsymbol{\alpha})|\,H\,|\Psi_T(\boldsymbol{\alpha})\rangle}{\langle\Psi_T(\boldsymbol{\alpha})|\Psi_T(\boldsymbol{\alpha})\rangle} \tag{22}$$

with respect to the parameters $\boldsymbol{\alpha}$.

Here we have hidden the complexity of providing good approximations to the many-body ground state in the work needed for a single evaluation of the function $E_V(\boldsymbol{\alpha})$. Indeed, the RHS of Eq. (22) contains two integrals that are in general very hard to be evaluated numerically:

$$E_V(\boldsymbol{\alpha}) = \frac{\langle\Psi_T(\boldsymbol{\alpha})|\,H\,|\Psi_T(\boldsymbol{\alpha})\rangle}{\langle\Psi_T(\boldsymbol{\alpha})|\Psi_T(\boldsymbol{\alpha})\rangle} = \frac{\int d^d r \Psi_T^\star(\mathbf{r};\boldsymbol{\alpha})\,H\Psi_T(\mathbf{r};\boldsymbol{\alpha})}{\int d^d r\,|\Psi_T(\mathbf{r};\boldsymbol{\alpha})|^2} \tag{23}$$

The success of VMC lies in its efficiency to evaluate these integrals using Monte Carlo techniques.

3.5.2 The variational principle for excited states

An important caveat to the above proof is that E_V is an *upper bound to the lowest eigenvalue whose eigenstate is not orthogonal to* $|\Psi_T\rangle$. This should be straightforward to see from Eq. (18) where if $|\Psi_T\rangle$ were orthogonal to $|\Phi_0\rangle$ then E_0 would be absent because it would be $a_0 = 0$. In that case we would get a bound for E_1 (similarly, if $\langle\Psi_T|\Phi_1\rangle \neq 0$. In the case were $|\Psi_T\rangle$ depends on variational parameters $\boldsymbol{\alpha}$, then the bound holds if for there is $\boldsymbol{\alpha}_0$ such that $\langle\Psi_T(\boldsymbol{\alpha}_0)|\Phi_0\rangle \neq 0$. Of course this hold in principle, in practice it's better if there is a finite (and preferably large) region in the $\boldsymbol{\alpha}$-space in which $|\Psi_T\rangle$ is not orthogonal to $|\Phi_0\rangle$. This is because in practice one needs to survey (at least some part of) the space of possible $\boldsymbol{\alpha}$'s to identify the minimum: isolate minima are hard to identify. This is where physical intuition is needed to identify the expected symmetries of the ground state and ensure that the trial state chosen has components that posses those symmetries.

While this caveat can be a nuisance when looking for the ground state, one can take advantage of it to use the variational principle to provide an upper bound for excited states of the Hamiltonian. This is done when by some physical principle we are certain of some symmetry of the ground state. Then we can construct a trial state $|\Psi_T\rangle$ that has no components with this symmetry guaranteeing that $\langle\Psi_T|\Phi_0\rangle = 0$

Note that for this to work then it should be $\langle\Psi(\boldsymbol{\alpha}|\Phi_0\rangle = 0$ *for all* possible values of $\boldsymbol{\alpha}$. This might sound trivial but we'll see in the example below how it can become a subtle point.

We will see this play out clearly in the example below.

3.5.3 Intuitive example from matrix diagonalization

In the above proof we have employed no physical argument and so the variational principle should hold for any matrix diagonalization: given a matirx O and a vector $\mathbf{v}$, the number $\mathbf{v}^\dagger O\mathbf{v}/\mathbf{v}^\dagger\cdot\mathbf{v}$ is greater or equal to the lowest eigenvalue of O that is not orthogonal to $\mathbf{v}$. We'll see this for the special case of a symmetirc 2×2 matrix.

Consider the symmetric matrix

$$O = \begin{pmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{pmatrix} , \tag{24}$$

with $O_{12} = O_{21}$. The diagonalization of this matrix is trivial and so we will use it as an example to get an insight on how the variational principle works. First we will diagonalize O by identifying the unitary transformation that makes it diagonal and then apply the variational principle to get an upper bound to the lowest eigenvalue. By doing so we'll see that if by some other considerations one knows enough about the symmetries of the lowest eigenvalue's vector, the variational principle can be invoked to identify it *exactly.* Let U be the unitary matrix that diagonalizes O, that is,

$$U^\dagger O U = \begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_2 \end{pmatrix} , \tag{25}$$

where ω_i are the eigenvalues of O.The requirement for U to be unitary provides three constraints for the elements of U:

$$U^\dagger U = \begin{pmatrix} U_{11}U_{11}+U_{21}U_{21} & U_{11}U_{12}+U_{21}U_{22} \\ U_{11}U_{12}+U_{21}U_{22} & U_{12}U_{12}+U_{22}U_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \implies \begin{array}{cl} U_{11}^2+U_{21}^2 & =1 \\ U_{11}U_{12}+U_{21}U_{22} & =0 \\ U_{12}^2+U_{22}^2 & =1 \end{array} \tag{26}$$

This means that U is described by a single parameter. It's then convenient to write U in terms of a parameter θ as

$$U(\theta) = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} \tag{27}$$

which satisfies all constraints and it's the most general form of an orthogonal matrix. Then it is

$$U^\dagger O U = \begin{pmatrix} O_{11}\cos^2\theta + O_{22}\sin^2\theta + O_{12}\sin 2\theta & \frac{1}{2}(O_{11}-O_{22})\sin 2\theta - O_{12}\cos 2\theta \\ \frac{1}{2}(O_{11}-O_{22})\sin 2\theta - O_{12}\cos 2\theta & O_{11}\cos^2\theta + O_{22}\sin^2\theta - O_{12}\sin 2\theta \end{pmatrix} \tag{28}$$

We can now choose $\theta = \theta_0$ such that this matrix becomes diagonal, that is, its off-diagonal elements, which are both the same, vanish, or

$$\frac{1}{2}(O_{11}-O_{22})\sin 2\theta_0 - O_{12}\cos 2\theta_0 = 0 \ . \tag{29}$$

Then the we can read the eigenvalues from the diagonal elements,

$$\omega_{\frac{1}{2}} = O_{11}\cos^2\theta + O_{22}\sin^2\theta \pm O_{12}\sin 2\theta \tag{30}$$

Note that this reproduces the known behaviour of the bifurcation of eigenvalues of a 2×2 symmetric matrix. The solution of Eq. (29,

$$\theta_0 = \frac{1}{2}\arctan\left(\frac{2O_{12}}{O_{11} - O_{22}}\right) \tag{31}$$

Further, we know that the columns of the transformation matrix U correspond to the eigenvectors and so

$$\mathbf{v}_1 = (\cos\theta_0, \sin\theta_0)^T \quad , \quad \mathbf{v}_2 = (\sin\theta_0, -\cos\theta_0)^T \quad , \tag{32}$$

and the diagonalization of O is done. Note that $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, as expected for the eigenvectors of a Hermitian matrix.

We will now pretend that we don't know the eigenvalues of O and apply the variational principle to get an upper bound to the lowest one. We choose for our $\mathbf{v}_T$ the normalized vector

$$\mathbf{v}(\phi) = (\cos\phi, \sin\phi)^T \quad . \tag{33}$$

Then we calculate

$$\omega(\phi) = \frac{\mathbf{v}(\phi)^\dagger O \mathbf{v}(\phi)}{\mathbf{v}(\phi)^\dagger \cdot \mathbf{v}(\phi)} = \sin^2(\phi)O_{11} - 2\cos(\phi)\sin(\phi)O_{12} + \cos^2(\phi)O_{12} \tag{34}$$

$$= \sin^2(\phi)O_{11} + \sin(2\phi)O_{12} + \cos^2(\phi)O_{12} \tag{35}$$

We can now minimize this function of a single variable by finding the roots of its derivative

$$\omega'(\phi) = 2\cos(2\phi)O_{12} - 2\cos(\phi)\sin(\phi)\left(O_{11} - O_{22}\right) \tag{36}$$

and solving for the root ϕ_0 gives

$$\omega'(\phi_0) = 0 \quad \Longrightarrow \quad \phi_0 = \frac{1}{2}\arctan\left(\frac{2O_{12}}{O_{11} - O_{22}}\right) = \theta_0 \tag{37}$$

That is we found the special case where the equal sign holds in Eq. (18). Now $\omega(\phi_0)$ is the lowest eigenvalue and $\mathbf{v}(\phi_0)$ is the best approximation of the corresponding eigenstate, which in this case its the eigenstate itself.

Given the above discussion now one might be tempted to use $\tilde{\mathbf{v}}(\phi) = (\sin\phi, -\cos\phi)$ to get the other eigenvalue since

$$\tilde{\mathbf{v}}^\dagger(\phi) \cdot \mathbf{v}(\phi) = 0 \tag{38}$$

for every ϕ. However, this would lead to the same ϕ_0. This is because

$$\tilde{\mathbf{v}}^\dagger(\phi) \cdot \mathbf{v}_1(\theta_0) = \sin(\phi)\cos(\theta_0) - \cos(\phi)\sin(\theta_0) \tag{39}$$

which is 0 only for $\phi = \theta_0$.
Try multiplying with $1/(\phi - \theta_0)$

3.5.4 Why Monte Carlo?

There is nothing in the Variational form of Scröndiger's equation that prohibits the use of your favourite technique for the evaluation of the integrals. One could try to evaluate Eq. (?) using e.g., Simpson's rule. When interested in one-particle in 1,2, or 3 dimensions, this might suffice. However, in the case of many-body systems, this would lead one to discover the curse of dimensionality: when using a quadrature rule whose power of the leading error is p, the integration error for a d dimensional integral scales with the number of function evaluations $\mathcal{N}$ we're willing to do as $\mathcal{N}^{-p/d}$. For a 3-dimensional system of N particles, $d = 3N$, and for Simpson's rule $p = 4$ leading to a scaling of $\mathcal{N}^{-1/(0.75N)}$. Then, even for a relatively small many-body system with $N = 10$, doubling the number of function evaluations divides the error by ≈ 1.01.

4 Diffusion Monte Carlo (DMC)

In the previous section we provided a robust way to produce an upper bound to the ground state energy. Variational Monte Carlo achieves this by minimizing the variational energy (calculated using Monte Carlo techniques) in a sub-space of the system's Hilbert space, which is parametrized by a set of variational parameters $\boldsymbol{\alpha}$. Depending on the choice of the functional form for the trial wavefunction the minimum of E_V (i.e., the tightest bound), can differ substantially from the ground state energy and approaching further might require substantial insight into the structure of the ground state, which we are looking for in the first place, or a prohibitively large number of variational parameters. In this sense, VMC is not easily systematically improvable.

In this section we will attempt a solution to this problem, that is, identifying the ground state energy, and properties of the ground state, through a clever recasting of Schröndiger's equation. We will describe what will turn out to be Diffusion Monte Carlo, which in principle, is systematically improvable and exact. Diffusion Monte Carlo, relies on the observation that Schröndiger's equation is a diffusion equation in imaginary time. Note that, up to here, we have been concerned with the time-independent Schröndiger's equation and this will remain the case; we're only using the time dependence as a trick to gain insight in the system's ground-state.

We start from the time-dependent Schröndiger's equation for a many-body state $|\Psi\rangle$,

$$i\hbar\partial_t |\Psi(t)\rangle = (H - E_T) |\Psi(t)\rangle \ , \tag{40}$$

where E_T is an arbitrary energy offset. Equation 40) has the general solution

$$|\Psi(t)\rangle = U(t) |\Psi(0)\rangle \tag{41}$$

where we defined the time evolution operator

$$U(t) = e^{i\frac{t}{\hbar}(H-E_T)} \tag{42}$$

making the substitution $t = i\tau$, we get

$$-\hbar\partial_\tau |\Psi\rangle = H |\Psi\rangle \ , \tag{43}$$

5 Auxiliary Field Diffusion Monte Carlo (AFDMC)